Rule v10: single-word factor-base selection, against its control

Run 2026-11-11, 20:26–23:30 UTC, local L0 (Intel Xeon @ 2.10 GHz, 4 vCPU, no PMU; `runs/host.json`). The protocol (PROTOCOL.md) was committed at `ef39d1e472`, before the change was implemented and before any row. Amendment 0 (`9d492639ce`) added the binaries and one implementation detail, also before the first row.

Outcome

- **P1 held at every size: the factor base and every logarithm are unchanged.** Every paired row has equal `identity_inputs` (which include the factor base's orbit representatives) and equal `counts` hashes, and recovers the same logarithms: 408 of 408 rows per run (`p1p2.json`). Both pins held against §23. In both analyses `logarithms_as_earlier` is 68 of 68 at every size, and `matched`'s `walks_checked` is 64 of 64.
- **Selection is 2.6–3.0x cheaper, short of P2's 4x.** The paired median of `select + select_projection` for `v10 / v10c` is 0.34–0.39. P2 asked for at most 0.25. It is **not met** at any size.
- **The cold distance to the matched rho fell at every size.** At n47 the paired cold ratio `v10 / v10c` is **0.851 [0.848, 0.855]**. That is inside P3's declared 0.81–0.86, so P3 is **met** there. At n57 it is **0.892 [0.876, 0.905]**, outside the declared 0.82–0.87, so P3 is **not met** there. The falsification bound (above 0.90 at n47) is not reached.
- **Distance remaining on this host.** The matched rho is still cheaper cold, by 3.75x [3.49, 4.07] at n47 (the control's distance is 4.41x), 4.43x at n57 (control 4.97x) and 13.64x at n61 (control 14.16x).

Under the owner's success criterion, this is a measured improvement [pcc:unverified] over the index-calculus baseline at all six sizes, with every answer checked. It is an engineering change to set-up. The algorithm and its counted work are unchanged.

What ran

change	`build_subgroup_orbit_factor_base_with_cost` takes a `u64` path for odd `n < 64` with an endomorphic Frobenius: one lift per Frobenius orbit, `MixHasher` tables and a single lowering. `projected_signed_orbit_map_fast` keeps its tables in `MixHasher`. Both are in `src/cryptanalysis/koblitz_index_calculus.rs`.
`v10`	`ic` from `f67d6c481b`, SHA-256 `deffa5b1…`
`v10c` (control)	`ic` from `ca1a767b1f` (the branch base, `main`), SHA-256 `07cd6558…`
harness	`icprog` from `f67d6c481b` (`V10`/`V10C`, `matched --comparison v10`). `isolated_bench` and `koblitz_rho_fixture` from `ca1a767b1f` (binaries.sha256).
sizes, targets	§23's six curves, 64 public targets and 4 A/A repeats a size, 408 rows a run. Seeds and parameter files are rule v9's.
order	manifests and pins first, then each size with `v10` before `v10c`
isolation	`isolated_bench run --wait --cpus 2`, `RAYON_NUM_THREADS=1`. The tool refused a start for load 37 times (`v10`) and 35 times (`v10c`). Six `v10` and four `v10c` attempts were contended and retried. Each kept row is the harness's first clean attempt.

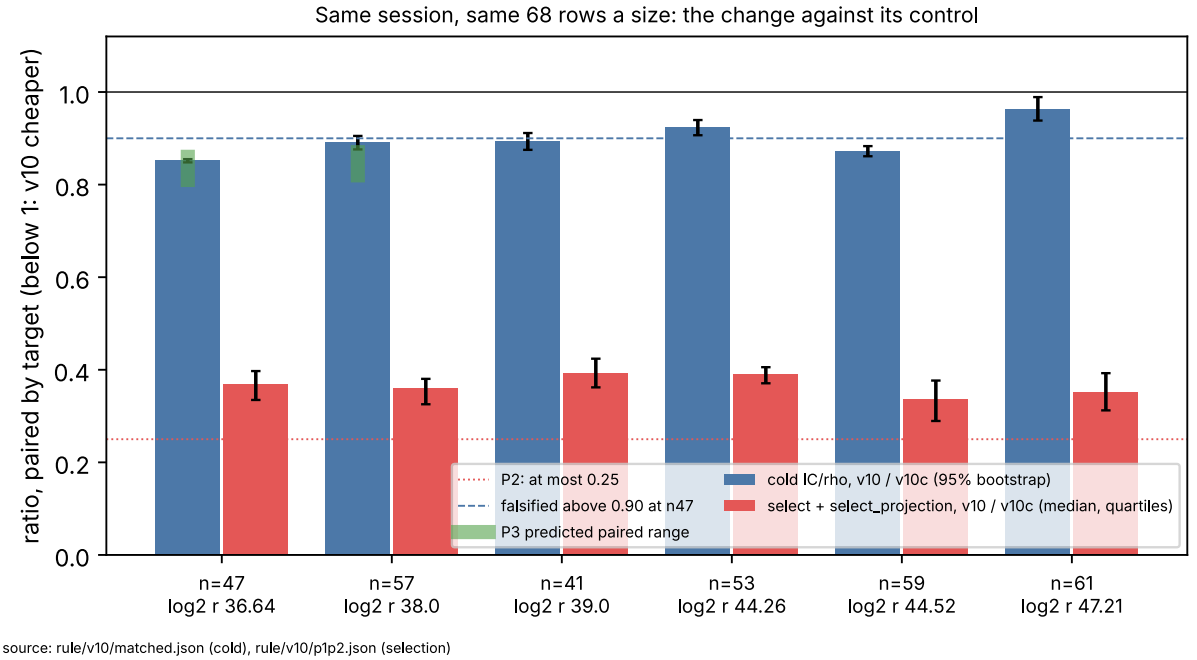
unit	in-process wall time in each row's own unit (`unit_ns`). The cold ratio is the IC's cold cost over its in-process matched rho's.

The steps are in `run.sh`. `claims` stopped once on a file outside the sparse checkout (`research/sat_factor_base_review_20260908/autolab/protocol.json`). Once that file was added, the remaining steps ran by hand in `run.sh`'s order (`run.log`). The bench record store (benchrec, S3) is not reachable from this container, so no benchrec record was opened. The raw run trees are committed as they are (29 MB and 28 MB).

Results

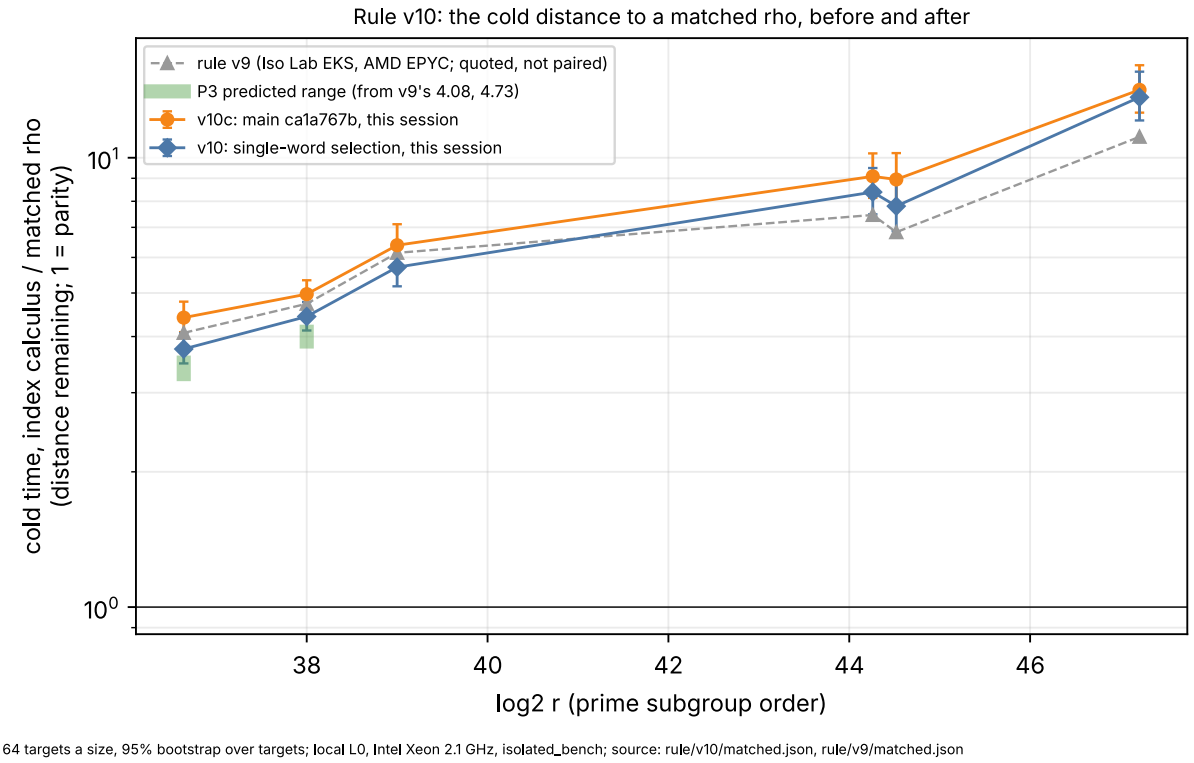
The cold figures are from `icprog rule matched --comparison v10` (`matched.json`; 64 targets, 95% bootstrap over targets, 10,000 resamples). The selection and set-up figures are from `p1p2.json`, made by `p1p2.sh` with jq. Each is the median over a row's repetitions, then the median over 68 rows. The brackets are the quartiles of the per-row ratio. The v9 column is quoted from rule v9 (Iso Lab EKS, AMD EPYC 9R45). It comes from another host and is not paired with these rows.

size	log2 r	v9 cold (EKS)	v10c cold	v10 cold [95%]	paired cold v10/v10c [95%]	select+proj ms v10c → v10	select ratio median [q25, q75]	set-up ms v10c → v10
n47 (a=1)	36.64	4.08	4.41	3.75 [3.49, 4.07]	0.851 [0.848, 0.855]	4.66 → 1.72	0.369 [0.335, 0.397]	20.5 → 17.8
n57 (a=0)	38.0	4.73	4.97	4.43 [4.13, 4.78]	0.892 [0.876, 0.905]	7.02 → 2.59	0.360 [0.325, 0.380]	34.9 → 30.7
n41 (a=0)	39.0	6.14	6.39	5.71 [5.17, 6.33]	0.894 [0.875, 0.911]	5.87 → 2.29	0.392 [0.362, 0.424]	49.4 → 45.7
n53 (a=0)	44.26	7.46	9.09	8.39 [7.48, 9.49]	0.923 [0.907, 0.939]	27.53 → 10.79	0.391 [0.371, 0.406]	359.8 → 336.8
n59 (a=1)	44.52	6.83	8.95	7.80 [6.87, 8.94]	0.872 [0.861, 0.883]	37.35 → 12.29	0.336 [0.289, 0.377]	421.3 → 368.8
n61 (a=0)	47.21	11.13	14.16	13.64 [12.11, 15.54]	0.963 [0.938, 0.989]	51.59 → 17.83	0.352 [0.312, 0.393]	1524.6 → 1467.7



source: rule/v10/matched.json (cold), rule/v10/p1p2.json (selection)

Paired ratios



64 targets a size, 95% bootstrap over targets; local L0, Intel Xeon 2.1 GHz, isolated_bench; source: rule/v10/matched.json, rule/v9/matched.json

Cold distance

Predictions against the outcome

	declared	measured	
P1	base, counters and logarithms identical	408/408 rows a run identical, pins held, logarithms 68/68 a size	**met**
P2	select+proj ratio ≤ 0.25	0.34–0.39 (medians)	**not met** at any size
P3 n47	paired cold 0.81–0.86 (3.3–3.5 of 4.08)	0.851 [0.848, 0.855]; 3.75 absolute here	**met** on the paired reading. The absolute reading cannot be compared across hosts.
P3 n57	paired cold 0.82–0.87 (3.9–4.1 of 4.73)	0.892 [0.876, 0.905]; 4.43 absolute here	**not met**
falsifier	paired cold at n47 > 0.90	0.851	not falsified

Notes on the reading

- **Why P2 fell short.** A Callgrind profile (development only, not a row) put the two phases at 9.0 M instructions against 32.5 M, a factor of 3.6. What remains is representation work that the base's `Vec<BinaryPoint>` format still requires: the projection's orbit walks and tables, lowering about 13,000 field elements into `BinaryPoint`s (`F2mElement::from_words`), and the packed orbit walk.
- **IC online time** was not targeted. Its paired ratio is 0.95–1.04 (`ic_online_matched_over_control`), and `v10` always ran first at each size. Set-up is about 99% of the cold cost, so this barely moves the cold ratio. It does show that a few percent of drift on the host within one size is inside these figures. No separate A/A run of two copies of one binary was made in this session; §23's four repeats a size are the in-run A/A.
- **The host.** This container's absolute ratios are higher than v9's EKS figures at every size, for the control as much as for `v10`. The v9 column is context only.
- **§5a.** Each run times both arms, the IC and its matched rho, in one process, so no native work is left unpriced. The 0, 0.1 and 1 GAE band gives one figure, and the order (rho cheaper cold at

every size) holds across it. The reference check found each run's rho no slower per step than the strong walk at every size (`rho_over_strong` 0.57–0.74).

What the validation gate would need

None of these was run here:

- The 2026-10-19 gate: a replay certificate and a validator report from `ic_validate`, like `rule/v9/replay-certificate.json` and `validator-report.json` (`src/bin/ic_validate.rs` would need a `v10` entry), plus `VALIDATION.json` and an independent review.
- An Iso Lab EKS (L2) rerun of this pairing, if Table A's rows are to move. Table A reads v9, which ran on EKS. These rows are local L0 on another host.
- A benchrec record with the raw tree in S3 (AGENTS.md §13a), and a `benchmarks/<date>-<slug>/` copy with a row in `benchmarks/README.md`, as the performance-results rule asks of the PR.

Next

Selection now costs about 0.35 of the control's. At n47 that is about 1.7 ms of 17.8 ms of set-up. The next set-up levers are `build` and `collect`, by size (from the rows' `setup_phases_ns`). Within selection, the remaining lever is to keep the base packed end to end instead of lowering every point.

Files

`PROTOCOL.md`, `run.sh`, `p1p2.sh`, `binaries.sha256` and `run.log`; the run trees `runs/` (v10) and `../v10c/runs/`; `claims/`, `manifests/` and `analysis.json` for each run; `matched.json` and `p1p2.json`; and `report/` (`plot.py`, the SVGs and PNGs, and `report.pdf`).